

SOVEREIGN VISION-BASED NAVIGATION

Autonomous X500 Geolocalization in GNSS-Denied Environments

1. THE PLATFORM: X500 EDGE-COMPUTE



- Drone Frame:** Holybro X500 V2 Quadcopter
- Flight Controller:** Pixhawk 6X running PX4 Autopilot
- Companion Computer:** Raspberry Pi 5 (Edge computer)
- Vision Sensor:** Raspberry Pi Camera Module 3 (Sony IMX708)
- Heading/Failsafe:** Here3 GNSS (DroneCAN protocol)
- Altitude Data:** Barometer feedback for altitude

2. SYSTEM INTEGRATION ARCHITECTURE

- Edge Sensor Capture**

The **Pi Camera 3** feeds raw video through the MIPI/CSI interface. To bypass native Ubuntu 22.04 limitations, custom **libcamera** drivers are invoked, granting direct, low-latency access to the Pi 5's Image Signal Processor (ISP).
- ROS 2 Processing Hub**

The Pi 5 processes visual data inside a **ROS 2 Humble** environment. The **cv_bridge** package translates ROS image topics into OpenCV matrices, feeding the dual-stage AI geolocalization algorithms running entirely on the edge CPU.
- High-Speed UART Bridge**

Standard 57600 baud rates caused packet drops ("ghosting nodes"). The system utilizes an upgraded **921600 baud rate** over **/dev/ttyACM0** to ensure uninterrupted, high-frequency MAVLink data transfer between the Pi 5 and the flight controller.
- Flight Control & EKF2 Fusion**

MAVROS translates the metric AI offsets and injects them directly into the **/mavros/vision_pose/pose** topic. The **Pixhawk 6X (PX4)** processes this via its EKF2 engine, issuing precise autonomous motor commands while strictly ignoring jammed GPS.

3. AI GEOLOCALIZATION PIPELINE



SIFT Feature Matching and Pixel Offset Calculation on Live Frame

Phase I: Semantic Global Retrieval

A quantized **ResNet-50** model extracts 512-dimensional embeddings from the live drone feed, comparing them against an indexed FAISS satellite database to find the closest semantic neighborhood map tile.

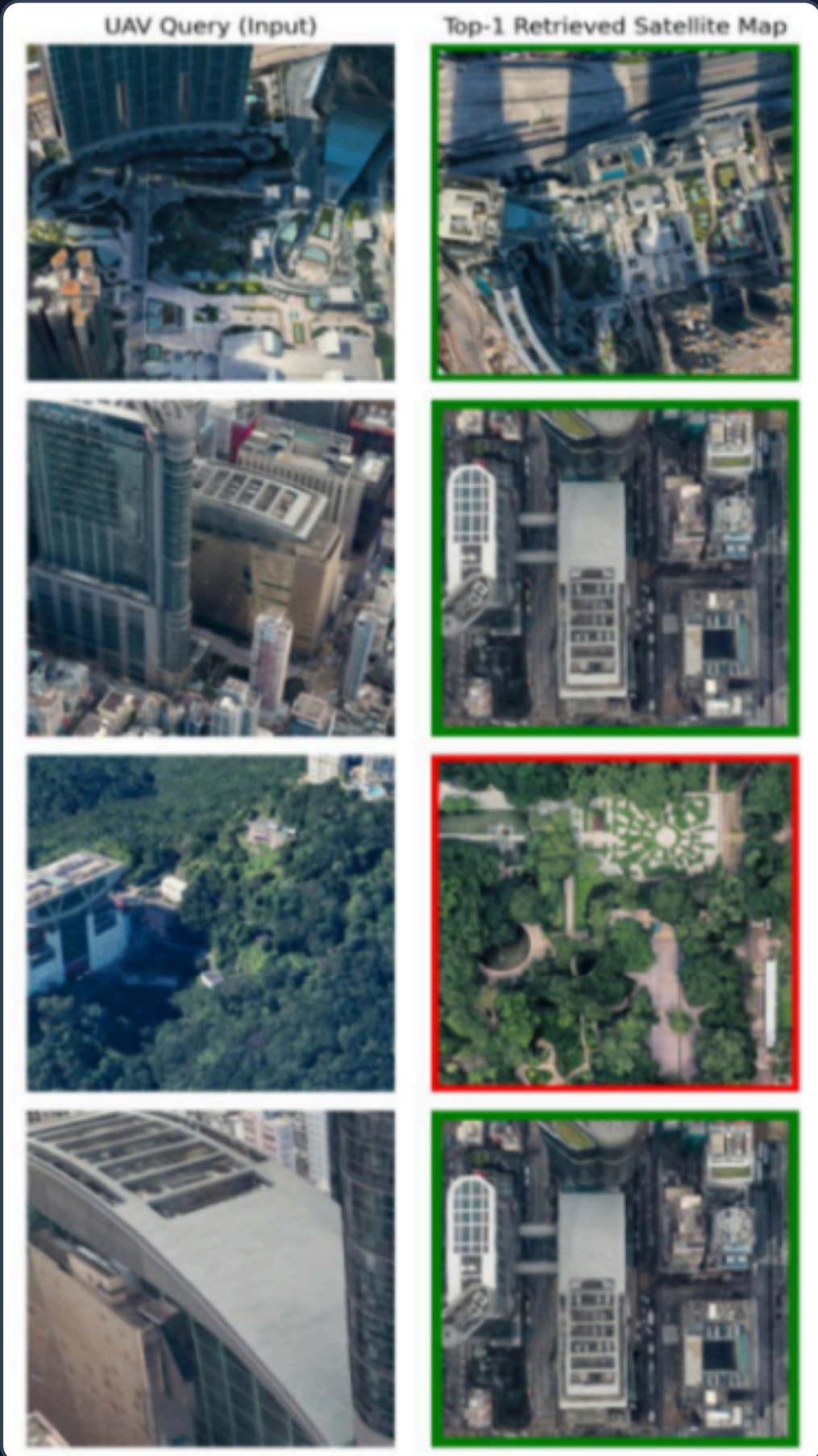
Phase II: Geometric Registration

The **SIFT** algorithm extracts distinct geometric keypoints (visualized above). **RANSAC** aggressively filters out mismatched or distorted points to ensure only high-confidence visual anchors are used to map the drone's location onto the tile.

Phase III: Dynamic Metric Translation

The system calculates **Ground Sample Distance (GSD)** using real-time altitude and camera FOV. Pixel offsets from Phase II are multiplied by the GSD to yield real-world metric X/Y corrections.

4. REAL-WORLD VALIDATION



Accuracy Verification: Drone view Vs satellite view AI model picks

Topographic Context (Amman): System successfully navigates aggressive elevation changes (Wadis, Jabals) and overcomes homogeneous limestone architecture ("texture traps").

Sovereign Operation: Provides complete resilience against regional GNSS jamming/spoofing threats by operating 100% locally on the Pi 5.

Proven Edge Performance: The optimized ONNX model successfully achieves a **62.5% Top-1 Retrieval Accuracy** while restricted solely to the computational limits of the edge CPU.